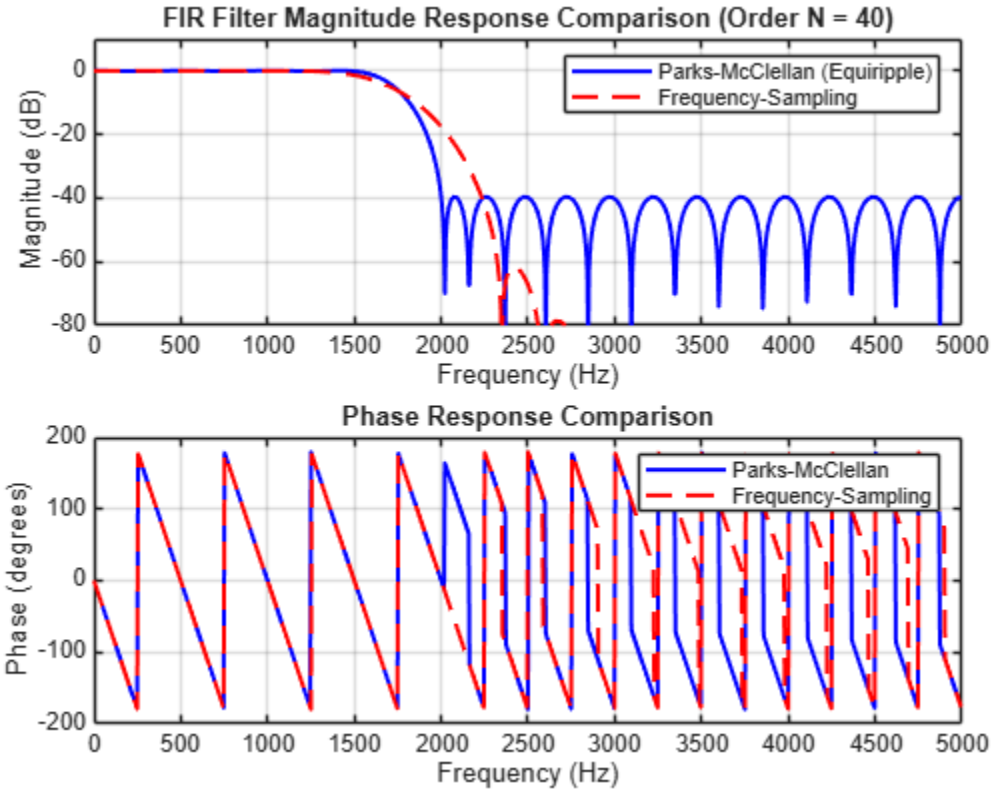
```
clear; clc; close all;

Fs = 10000;
fp = 1500;
fs = 2000;
N = 40; % Filter order

% 1. Parks-McClellan (Equiripple) Design
f_bands = [0, fp, fs, Fs/2] / (Fs/2); % Normalized frequency band edges
a_bands = [1, 1, 0, 0];               % Desired amplitudes
b_pm = firpm(N, f_bands, a_bands);

% 2. Frequency-Sampling Design
b_fs = fir2(N, f_bands, a_bands);

% ========================================
% DISPLAY NUMERICAL OUTPUTS IN COMMAND WINDOW
% ========================================
fprintf('=== EXPERIMENT 5: PARKS-MCCLELLAN vs FREQUENCY SAMPLING ===\n\n');
fprintf('Filter Order (N): %d\n', N);
fprintf('Passband Edge: %d Hz, Stopband Edge: %d Hz\n\n', fp, fs);

fprintf('--- PARKS-MCCLELLAN COEFFICIENTS (b_pm, First 10) ---\n');
disp(b_pm(1:10));

fprintf('--- FREQUENCY-SAMPLING COEFFICIENTS (b_fs, First 10) ---\n');
disp(b_fs(1:10));

% Frequency Responses
[H_pm, f] = freqz(b_pm, 1, 1024, Fs);
[H_fs, ~] = freqz(b_fs, 1, 1024, Fs);

% Plotting
figure('Name', 'Parks-McClellan vs Frequency-Sampling');
subplot(2,1,1);
plot(f, 20*log10(abs(H_pm)), 'b', 'LineWidth', 1.5); hold on;
plot(f, 20*log10(abs(H_fs)), 'r--', 'LineWidth', 1.5); grid on;
legend('Parks-McClellan (Equiripple)', 'Frequency-Sampling');
title(['FIR Filter Magnitude Response Comparison (Order N = ', num2str(N),
')']);
xlabel('Frequency (Hz)'); ylabel('Magnitude (dB)'); ylim([-80 10]);

subplot(2,1,2);
plot(f, angle(H_pm)*180/pi, 'b', 'LineWidth', 1.5); hold on;
plot(f, angle(H_fs)*180/pi, 'r--', 'LineWidth', 1.5); grid on;
legend('Parks-McClellan', 'Frequency-Sampling');
title('Phase Response Comparison');
xlabel('Frequency (Hz)'); ylabel('Phase (degrees)');
clear; clc; close all;

Fs = 10000;
fp = 1500;
```

```
fs = 2000;
N = 40; % Filter order

% 1. Parks-McClellan (Equiripple) Design
f_bands = [0, fp, fs, Fs/2] / (Fs/2); % Normalized frequency band edges
a_bands = [1, 1, 0, 0];               % Desired amplitudes
b_pm = firpm(N, f_bands, a_bands);

% 2. Frequency-Sampling Design
b_fs = fir2(N, f_bands, a_bands);

% ========================================
% DISPLAY NUMERICAL OUTPUTS IN COMMAND WINDOW
% ========================================
fprintf('=== EXPERIMENT 5: PARKS-MCCLELLAN vs FREQUENCY SAMPLING ===\n\n');
fprintf('Filter Order (N): %d\n', N);
fprintf('Passband Edge: %d Hz, Stopband Edge: %d Hz\n\n', fp, fs);

fprintf('--- PARKS-MCCLELLAN COEFFICIENTS (b_pm, First 10) ---\n');
disp(b_pm(1:10));

fprintf('--- FREQUENCY-SAMPLING COEFFICIENTS (b_fs, First 10) ---\n');
disp(b_fs(1:10));

% Frequency Responses
[H_pm, f] = freqz(b_pm, 1, 1024, Fs);
[H_fs, ~] = freqz(b_fs, 1, 1024, Fs);

% Plotting
figure('Name', 'Parks-McClellan vs Frequency-Sampling');
subplot(2,1,1);
plot(f, 20*log10(abs(H_pm)), 'b', 'LineWidth', 1.5); hold on;
plot(f, 20*log10(abs(H_fs)), 'r--', 'LineWidth', 1.5); grid on;
legend('Parks-McClellan (Equiripple)', 'Frequency-Sampling');
title(['FIR Filter Magnitude Response Comparison (Order N = ', num2str(N),
')']);
xlabel('Frequency (Hz)'); ylabel('Magnitude (dB)'); ylim([-80 10]);

subplot(2,1,2);
plot(f, angle(H_pm)*180/pi, 'b', 'LineWidth', 1.5); hold on;
plot(f, angle(H_fs)*180/pi, 'r--', 'LineWidth', 1.5); grid on;
legend('Parks-McClellan', 'Frequency-Sampling');
title('Phase Response Comparison');
xlabel('Frequency (Hz)'); ylabel('Phase (degrees)');
```

```
=== EXPERIMENT 5: PARKS-MCCLELLAN vs FREQUENCY SAMPLING ===

Filter Order (N): 40
Passband Edge: 1500 Hz, Stopband Edge: 2000 Hz

--- PARKS-MCCLELLAN COEFFICIENTS (b_pm, First 10) ---
  Columns 1 through 7

   -0.0015    0.0072    0.0053   -0.0010   -0.0081   -0.0072    0.0037
```

```
  Columns 8 through 10

    0.0142    0.0100   -0.0090

--- FREQUENCY-SAMPLING COEFFICIENTS (b_fs, First 10) ---
  Columns 1 through 7

    0.0000    0.0001    0.0002   -0.0001   -0.0008   -0.0009    0.0008

  Columns 8 through 10

    0.0035    0.0029   -0.0039

=== EXPERIMENT 5: PARKS-MCCLELLAN vs FREQUENCY SAMPLING ===

Filter Order (N): 40
Passband Edge: 1500 Hz, Stopband Edge: 2000 Hz

--- PARKS-MCCLELLAN COEFFICIENTS (b_pm, First 10) ---
  Columns 1 through 7

   -0.0015    0.0072    0.0053   -0.0010   -0.0081   -0.0072    0.0037

  Columns 8 through 10

    0.0142    0.0100   -0.0090

--- FREQUENCY-SAMPLING COEFFICIENTS (b_fs, First 10) ---
  Columns 1 through 7

    0.0000    0.0001    0.0002   -0.0001   -0.0008   -0.0009    0.0008

  Columns 8 through 10

    0.0035    0.0029   -0.0039
```



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